

JKBOSE 2026

CLASS 11TH

Anatomy Of Flowering Plants

LECTURE-04

BOTANY

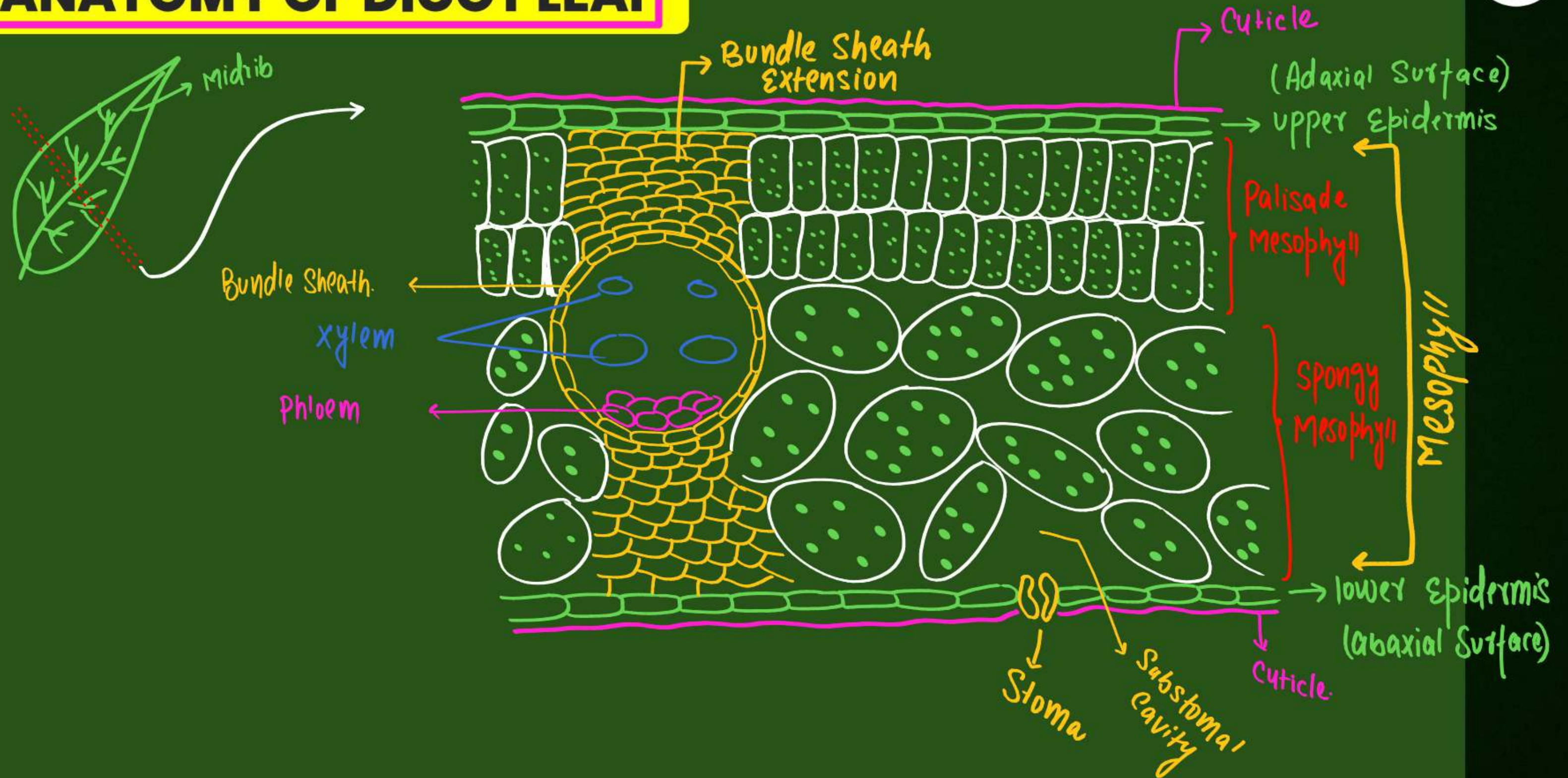
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Topics To Covered

- 1** ANATOMY OF DICOT LEAF
- 2** ANATOMY OF MONOCOT LEAF
- 3** ...

ANATOMY OF DICOT LEAF



⇒ **EPIDERMIS:** It covers both upper and lower surfaces, upper epidermis possess less or no stomata, lower surface possess more stomata, Epidermis possess thick cuticle.

⇒ **MESOPHYLL:** Tissue present between upper and lower epidermis, possesses chloroplast, parenchymatous, performs photosynthesis.

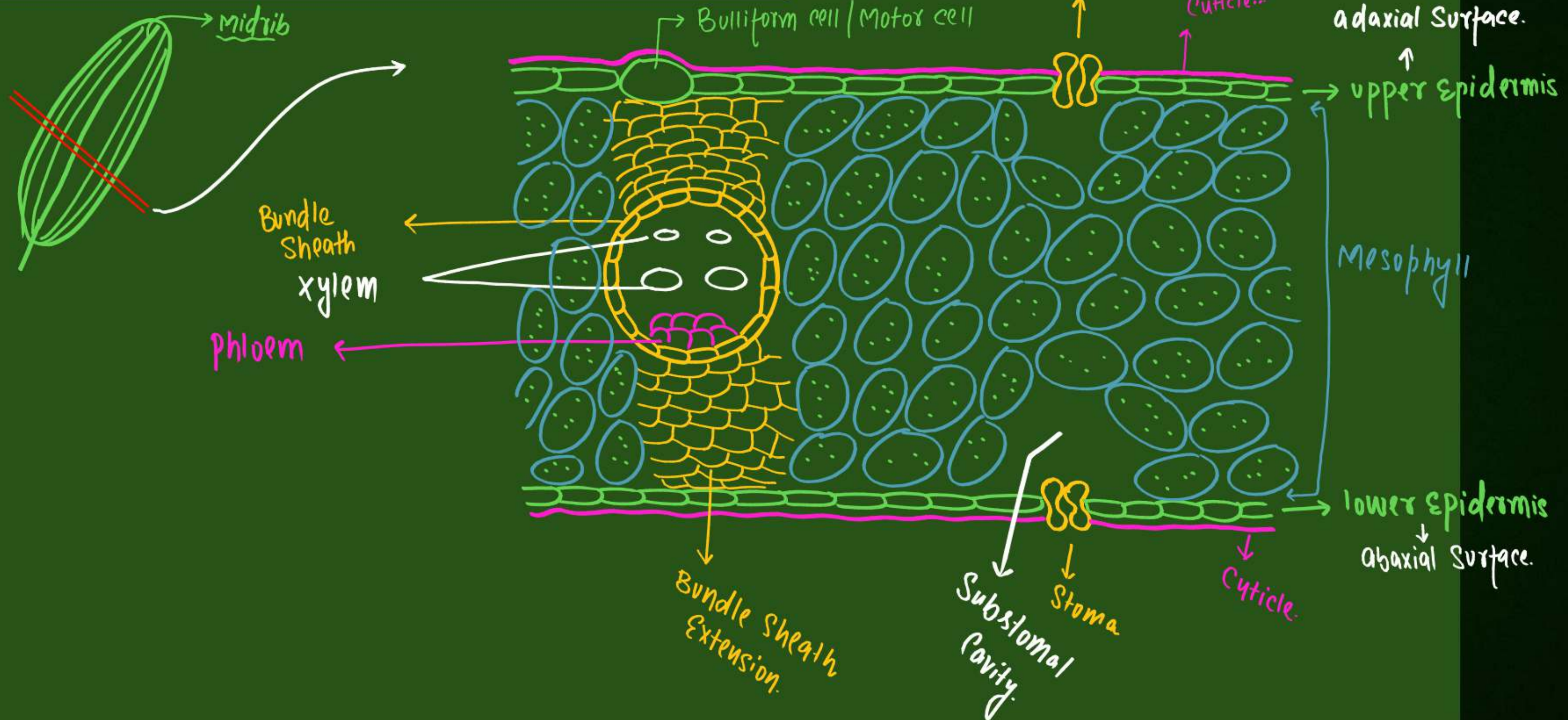
⇒ It has two types of cells:

- a) **Palisade parenchyma-** towards adaxial surface, elongated, arranged vertically, parallel to each other.
- b) **Spongy parenchyma-** towards abaxial surface, oval/round, loosely arranged, possess air spaces and air cavities.

➡ **VASCULAR BUNDLE:** seen in veins and midrib.

- Size of VB depends on size of veins. In reticulate venation veins vary in thickness. Vascular bundles are surrounded by layer of thick-walled cells called as bundle sheath cells.
- Vascular bundles – conjoint, collateral, closed.
- Xylem towards upper surface and phloem towards lower surface.

ANATOMY OF MONOCOT LEAF



⇒ **EPIDERMIS:** It covers both upper and lower surface.

- Almost equal number of stomata are present on both surfaces (isostomatic).
- Cuticle covers both the surfaces.
- On upper epidermis, bulliform cells / motor cells are present. These cells are large, empty, colourless.
- Due to water absorption, these cells become turgid & makes the leaf exposed.
- Due to water loss, these cells become flaccid & makes the leaf to curl inwards to minimize water loss.

⇒ **MESOPHYLL:** parenchymatous, present between two epidermal layers, not differentiated into palisade and spongy, possess chlorophyll, performs photosynthesis.

↳ VASCULAR BUNDLE:

- Thickness of vascular bundles are similar due to parallel venation (except in midrib).
- Surrounded by bundle sheath.
- Vascular bundle --- conjoint and closed.

QUESTIONS

Given below are two statements:

Statement-I: In a dicotyledonous leaf, the adaxial epidermis generally bears more stomata than the abaxial epidermis.

False.

Statement-II: In a dicotyledonous leaf, the adaxially placed palisade parenchyma is made up of elongated cells, which are arranged vertically and parallel to each other.

True

In the light of the above statements, choose the correct answer from the options given below:

(2024 Re)

- A** Statement-I is true but Statement-II is false
- B** Statement-I is false but Statement-II is true
- C** Both Statement-I and Statement-II are true
- D** Both Statement-I and Statement-II are false

QUESTIONS

Large, empty colourless cells of the adaxial epidermis along the veins of grass leaves are
(2020 Covid)

- A** Guard cells
- B** Bundle sheath cells
- C** Bulliform cells
- D** Lenticels

QUESTIONS

Bulliform cells are responsible for

(2024)

- A** Increased photosynthesis in monocots.
- B** Providing large spaces for storage of sugars.
- C** Inward curling of leaves in monocots.
- D** Protecting the plant from salt stress.

QUESTIONS

Grass leaves curl inwards during very dry weather. Select the most appropriate reason from the following (2019)

- A** Closure of stomata
- B** Flaccidity of bulliform cells
- C** Shrinkage of air spaces in spongy mesophyll
- D** Tyloses in vessels

QUESTIONS

What is true about a monocot leaf?

(1992)

- A** Reticulate venation ^x
- B** Absence^x of bulliform cells from epidermis
- C** Mesophyll not differentiated into palisade and spongy tissues
- D** Well differentiated mesophyll

HOMEWORK

Revision



Complete Chapter.

Thank
You